



Implicit neural representations for surrogate modeling in the built environment

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ABSTRACT

Physical phenomena, including aerodynamics and heat transfer, exhibit complex shape-dependent relationships with building geometry, shaping microclimates that directly affect urban livability and comfort. In design and engineering practice, surrogate models reduce the computational burden of simulations, providing faster and more iterative performance feedback within design workflows. This paper introduces Per-FORM, a framework that leverages implicit neural representations (INRs) for predictive modeling in the built environment. The approach accommodates variations in geometric complexity, scale, and topology while representing continuous physical fields through decoupled modules encoding both geometry and building influence. Its ability to infer full-field and near-surface predictions is evaluated across multiple metrics, demonstrating state-of-the-art accuracy for complex geometries. Beyond predictive accuracy, Per-FORM brings simulation-in-the-loop feedback into digital workflows, supporting performance-informed exploration, ideation, and conceptualization, and enriching informed creative processes in design and engineering practice.

1. Introduction

Physics-based simulations play a fundamental role in enabling informed decision-making across design and engineering fields. In the context of the built environment, they enable the evaluation of the aerodynamics, heat exchanges and microclimate fluctuations that characterize the complex airflow and thermal patterns on and surrounding buildings [1]. These are crucial for determining the potential consequences of design, planning and policy decisions on the livability and comfort of urban residents as well as the repercussions of inaction in the face of a warming climate. The computational complexity associated with carrying out such simulations remains largely prohibitive to their smooth and practical integration into fast-paced iterative computer-aided design (CAD) workflows, which are central to advancing sustainable built environments. This has motivated decades of research into surrogate modeling, a field dedicated to constructing approximate models that emulate the behavior of high-fidelity simulations. Capturing the complexities and irregularities of 3D data for processing into deep learning pipelines remains, however, challenging and current approaches are predominantly topology-, modality- and discretization-dependent (refer to Section 2.1 for a review on 3D representations for surrogate models). This limits their ability to handle variations in geometric scale and complexity, and their efficacy in representing continuous physical fields.

Implicit neural representations (INRs) have been recently leveraged as an alternative representation that can efficiently encode 3D shapes by fitting continuous functions using coordinate-based networks [2–4]. This enables the decoupling of discretization resolution and modality from the complexity of the representation, providing a unified and scalable solution, not just for single-shape high-quality reconstructions [5], but also extends to learning parametrizations for entire classes of shapes [2,6] and multi-modal inputs [7]. Their potential thus extends far beyond the dominant focus on reconstruction (refer to Sections 2.2, 2.3 for a review on INRs as functional representations — methods and applications), but the viability of their broader adoption in surrogate modeling of physics fields remains an emerging research area and has yet to be fully investigated in the context of large-scale and complex built environments, where accelerated and sustainability-informed design processes are increasingly essential.

This paper developed Per-FORM, a framework that leverages INRs for the coupled representation and predictive modeling of computationally prohibitive and/or non-differentiable geometric and physical quantities for the built environment. Per-FORM establishes a mapping between shape-conditioned compact learned encodings of geometric data and their associated physical fields. In contrast to competing approaches, this process enables field and near-surface performance inference for arbitrary geometries, while also defining seamless forward and inverse correspondences between shape and performance,

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enabling streamlined integration of performance feedback into design and engineering processes.

The key contributions of this paper are:

- **An INR-based surrogate modeling framework for full flow field prediction in the built environment, with broader applicability to complex geometries:** Per-FORM adapts INRs to surrogate modeling in a domain characterized by large physical extent, fine-scale features, and high topological complexity. While demonstrated on buildings, the formulation is general and structurally extensible to other application areas where existing INR work (e.g. automotive, airfoil) has so far been limited to more constrained topologies. The framework achieves competitive performance relative to grid-based surrogates in the built environment, while broadening the representational scope of both geometry (from extrusion models to 3D forms) and predictive output (from single-plane to full-field and near-surface predictions).
- **Beyond just surrogate modeling, leveraging the structure of the learned performance latent space for performance-informed design:** Beyond enabling the integration and acceleration of physics field inference for building applications, Per-FORM demonstrates that the structure of the learned performance latent space can directly support performance-informed design tasks. This is exemplified through design applications such as performance-based categorization of building morphologies and performance-informed design ideation, illustrating new pathways for embedding performance feedback into computer-aided design processes and accelerating performance-informed design workflows.

The paper, additionally, develops and compares three instantiations of the framework that explore the role of spatial discretization of latent embeddings in capturing geometric complexity and leveraging spatial structure. These variants demonstrate improved performance at reduced computational cost in domain-specific evaluations. While compositional modeling is not explored in this work, the framework's latent representation scheme structurally supports such extensions through spatially localized embeddings, aligning with prior findings on multi-object representation learning. At the core of this work is the effort to address three central questions: (1) Can implicit neural representations provide a tractable and scalable surrogate formulation for built environment physics? (2) How effectively can such representations encode coupled geometric and physical relationships for joint field and surface predictions? (3) In what ways can the learned representation be leveraged to support performance-informed design and generalization across diverse geometries?

2. Related work

Research on surrogate modeling and implicit representations spans a broad spectrum of spatial formulations and learning frameworks. From grid-based convolutional surrogates to graph-, point-, and operator-based models, each approach presents distinct trade-offs between geometric flexibility, computational cost, and representational continuity. This section situates the proposed framework within that broader context, outlining major 3D representation strategies, their application in the built environment, and recent developments in implicit neural representations for physics-informed learning.

2.1. 3D representations for surrogate models

The computational complexity associated with physics-based simulations can be prohibitive to their smooth and practical integration into fast-paced iterative design and engineering workflows. This has motivated decades of research into surrogate modeling, in particular deep learning due to their remarkable capability to learn high-dimensional,

non-linear, and complex phenomena [8,9]. Existing surrogate modeling frameworks of physics fields rely primarily on grid-based approaches using convolutional neural networks (CNNs) [10–13], due to their strong ability to capture local and multi-scale relationships but they struggle to represent irregular geometries and high-resolution domains as they become intractable at higher dimensions [14]. These have been successfully extended to unstructured domains using architectures directly operating on point clouds [15–17] but these are unable to capture edge and pairwise relationships between nodes constraining them to simple geometries [18], and graph neural networks (GNNs) [19–21] which more effectively handle irregular geometries but are limited in their receptive field and information propagation particularly in large graphs [22] and cannot generalize across mesh topologies and levels of discretization [23]. Transformer-based surrogates [24,25] have recently emerged, operating on enriched point sets augmented with implicit encodings (e.g., SDFs, normals). While offering strong representational capacity, they require dense inputs to resolve geometric detail and produce fields via interpolation rather than direct inference, introducing resolution dependence and potential smoothing artifacts. The inherent topology-, modality- and discretization-dependence of these surrogate modeling approaches limit their ability to handle variations in geometric scale and complexity, and their efficacy in representing continuous physical fields.

2.1.1. Built environment domain-specific applications

In the context of the built environment, grid-based convolutional approaches have been widely adopted as the state-of-the-art method for surrogate modeling of physics fields on and around buildings including wind velocity fields, wind pressure coefficients, solar radiation among other applications. Earlier work recorded substantial speed-ups at low accuracy trade-offs based on low-resolution 3D voxel grids [26] and 2.5D height map representations [27,28]. This has resulted in a large body of research that followed, dedicated to adopting, testing and improving these UNet-based approaches primarily utilizing variations of the 2.5D encoding strategy [29–37]. These variants include augmentations with positional encodings like signed distance fields [38] or integration of building subtractive elements, trees and topography by using additional encoding channels [39] or multi-channel strategies for height encodings [40]. Mesh-, graph-, and point-cloud-based representations are effective in compact-object aerodynamics (e.g., cars, airfoils), where domains typically involve 10^3 – 10^5 nodes [41–43], and neural PDE solvers rarely scale beyond 4×10^5 – 7×10^5 points [25]. Even state-of-the-art datasets and models extended to million nodes [25, 44] remain orders of magnitude below the discretizations required in the built environment. Unlike compact objects with tightly bounded flow domains, building simulations must capture both detailed surface geometry and the entire surrounding field, pushing meshes to 5×10^5 – 10^7 cells for single buildings and can scale up to 10^8 for neighborhoods with meter-level resolution. Downsampling to fit existing frameworks inevitably sacrifices fidelity, degrading both surface metrics (e.g., façade pressures) and field-level outcomes (e.g., pedestrian comfort). These scale requirements make direct learning on full meshes, graphs, or point clouds intractable, as built-environment domains routinely span tens to hundreds of millions of elements. While a few studies have investigated outdoor flow prediction using graph neural networks (GNNs), physics-informed neural networks (PINNs), or neural operator (NOs) models [45–47], their use has been largely restricted to tasks such as continuous field inference from sparse sensor observations or short-horizon prediction in transient simulations, rather than scalable, geometry-aware inference across diverse building morphologies and configurations. Table 1 summarizes the major surrogate modeling approaches, outlining their core representations, computational characteristics, and scalability limits, and situating the implicit formulation adopted in this work within the broader landscape of existing 2D, 3D, point-, mesh-, and operator-based methods.

Table 1

Comparison of surrogate modeling architectures for physics-based prediction.

Method	Representation	Field dim.	Cont.	Pred.	Const.	Depend.	Mem. scaling	Typical limit	Geom. opt.
2D CNN	Raster grid	2.5D	D	FS	R	F	$\mathcal{O}(N^2)$	512^2 – 1024^2	N
3D CNN	Voxel grid	3D	D	F	R,T	F	$\mathcal{O}(N^3)$	64^3 – 128^3	N
Point-based	Unstructured points	3D	D	S,PF	R	PF	$\mathcal{O}(N)$	$\leq 10^5$ points	ID
GraphNN	Mesh/graph nodes	3D	D	S,PF	T	PF	$\mathcal{O}(N \log N)$	$\approx 3 \times 10^5$ nodes	ID
Neural Operator	Continuous coordinates	2D–3D	C	F	–	PF	$\mathcal{O}(N-N^2)$	$\leq 10^5$ samples	N
Transformer	Tokens/patches	3D	D	F,PF	R	F	$\mathcal{O}(N^2)$	$\leq 64^3$ tokens	N
INR	Coordinates + latent embedding	3D	C	S,F	–	–	$\mathcal{O}(N)$	Arbitrary (query-based)	D

Abbreviations. *Field dim.* = Field dimensionality (2D, 2.5D, 3D). *Cont.* = Field continuity (D = Discrete, C = Continuous). *Pred.* = Predicted field coverage (S = Surface, F = Field, FS = Field slice, PF = Partial field). *Const.* = Geometry constraint (R = Resolution, T = Topology). *Depend.* = Domain dependency (F = Fixed, PF = Partially fixed/shared grid, – = Domain-agnostic). *Mem. scaling* = Computational memory complexity with number of samples N . *Geom. opt.* = Geometry-aware optimization capacity (N = None, ID = Indirect/template-based deformation, D = Direct/learned implicit embedding).

2.2. INRs as functional representations

Implicit neural representations (INRs) have been recently leveraged as an alternative representation that can efficiently encode 3D shapes by fitting continuous functions using coordinate-based networks [2–4]. This enables the decoupling of discretization resolution and modality from the complexity of the representation, providing a unified and scalable solution, not just for single-shape high-quality reconstructions [5], but also extends to learning parametrizations for entire classes of shapes [2,6]. They have as well been explored in hybrid settings, through spatially-discretized latent vectors, in the form of feature voxel grids [48–51], hierarchical octrees [52,53], ground plan [54], triplanar factorizations [51,55] among others. These have been widely adopted to handle higher geometrical complexity (e.g. multi-objects scenes) at a lower computational network complexity. INRs have also been extended to represent a wide range of modalities such as images [56], videos [57], sound [58], medical imaging [59,60], climate data [61], architectural and urban environments [62–64], and physical fields [65,66]. Beyond functioning as continuous discretization-agnostic representations, INR applications extend to include generative modeling [67], and more recently have been adopted as functional inputs to other more general downstream deep learning tasks such as classification [67] and segmentation [68]. Interest in extending the usability of INRs to the physics domain is an emerging area of research, including applications such as inference of physical attributes from videos [69,70], simulation of objects dynamics based on physical properties [71–73], reduced-order modeling of physics fields [74] and topology optimization [75].

2.3. Deep learning on INRs

Recent research has expanded implicit neural representations from pure function fitting toward task-driven learning. Through coupling INRs with differentiable encoders and decoders, deep learning frameworks now treat these functions as transferable representations, linking continuous geometry to higher-level reasoning and prediction tasks.

2.3.1. Learning the INR functional embedding

Learning INR shape encodings has been explored through two distinct approaches: a weight-encoded strategy that fits an individual network for each shape [5], and a globally conditioned embedding strategy for each shape that shares the same base network weights [2]. While the first approach has been shown to produce superior high-quality reconstructions on complex shapes [5,7,76] and its deployment is not reliant on the availability of a complete dataset, it suffers from a much higher computational and storage overhead [77]. Converting these weight parametrizations into compact and meaningful embeddings is additionally non-trivial due to the high dimensionality of the weight-space [78], and direct mapping of weights across different networks is not possible due to the randomness involved in training [77]. Adopting a shared-network base coupled with shape-conditioned latent embeddings, which is the second approach and the one adopted in

this work, provides a more straightforward strategy for downstream learning approaches which is to directly process its compact latent vectors as inputs for other networks. This conditioning has been implemented through auto-decoding [2], globally defined and spatially discretized latent and shift modulations [67,79], hypernetworks [61, 80] as well as integrated in modality-specific architectures including auto-encoders [68]. In the auto-decoding approach, adopted in this work, initially randomized latent codes for individual shapes are optimized concurrently with the network weights during training, and during inference, with frozen network weights, newly initialized latent codes are fit to unseen shapes.

2.3.2. Integrating the INRs into downstream applications

Embedding INRs for task-specific applications has been approached in three distinct strategies: coupled, decoupled, and contextualized. The first approach is through directly learning task-specific metrics or jointly learning the representation and its performative attributes [61, 81,82], which embeds non-representational characteristics in the learning but comes at the cost of a trade-off in reconstruction quality and would not be suitable for any other application without re-training. To enable a more general usability of the INR, others proposed to de-couple the representation learning from downstream tasks through using the latent vectors as frozen inputs to other networks [67,77–79], or contextualizing them by concatenating with fine-tuned learned features aimed at recapturing shape-specific information meaningful for the task at hand [68]. This work builds on the existing decoupled and contextualized approaches to enable the usability of the geometry INRs to be used across various performance downstream applications. The coupling of global latent representations with MLP or transformer-based field decoders for downstream deep learning tasks has been shown to be successful in extracting features for simple problems [67] but their expressive abilities struggled to perform in par with other state-of-the-art approaches as the dataset and problem complexity increased [67,79]. Spatial discretization of the latent embeddings, which when coupled with local conditioning of a field decoder, expands the model's capacity to capture higher geometrical complexities at lower computational costs [51], has also been shown to be advantageous for the downstream usability of the INRs on more challenging tasks that may benefit from powerful architectures that can exploit spatial structures [79,83]. These findings motivated this work's investigation into the role of latent discretization for the yet unexplored context of physics surrogate models.

2.3.3. INRs for geometry-aware inference of physics fields

In the surrogate modeling context of physics flows, INRs have been primarily explored for dynamics modeling [66,84–87], solving sets of differential equations over time within pre-defined boundary conditions, as well as continuous field reconstruction from sparse observations [65,88–90]. A limited body of work [18,22,91–93] addresses the application of geometry-aware inference of physical fields, which is fundamental to design and optimization domains. This is accomplished through a framework that involves learning distinct embeddings for

geometry and physical fields, along with a mapping between them. This approach is analogous to the contextualized INR technique used for downstream learning [68] and neural operators [94] that are increasingly applied in deep learning for scientific simulations. Among these, Geo-FNO [91], Geom-DeepONet [18], CORAL [92] adopt learned mesh-deformations for encoding shape variations, [95]: GNN-based embeddings, [93]: pre-defined embedding based on engineered design parameters, and only [23] and INFINITY [22] capitalize on learned INR shape embeddings. The approach taken in this work is the latter, as it is uniquely capable of representing any arbitrary geometry, without being constrained by topology or requiring handcrafted parametric spaces. In contrast to INFINITY [22] which was applied to surface and field inference of 2D airfoil designs and [23] on surface inference for 3D aircraft wings, the goal of this work is to develop a surrogate model for 3D buildings which introduces additional challenges, including greater variations in morphology, topology and scales.

3. Surrogate performance modeling on implicit neural representations

Surrogate modeling on implicit neural representations extends the use of INRs beyond geometry reconstruction toward coupled inference of physical behavior. By representing both shape and performance as continuous functions over spatial coordinates, the framework enables direct mapping between geometric form and physical response.

3.1. Buildings as a class of objects

Buildings, as the dominant class of design elements in the built environment, present unique challenges: (1) *Scale*: Buildings, in contrast with common classes of objects in product and industrial design fields, vary substantially in scale from a single-story building to mixed-use complexes and skyscrapers reaching hundreds of meters in span or height. The spatial resolution of interest for performance predictions is a factor of an absolute (not a relative) scale and is generally in the order of the human scale. Most environmental performance metrics are as well scale-dependent. (2) *Orientation*: The physical influence of buildings is orientation-dependent as it is, for most environmental performance metrics, primarily a resultant of climatic parameters such as sun angles and wind direction. (3) *Morphology*: x-y plane morphological variations are more pronounced than variations along the z direction for most, but not all, buildings, making ground plans and layouts base common (but not exclusively sufficient) representations of buildings in the architectural design field. (4) *Influence*: Buildings' physical influence on the surrounding environment extends far beyond the surface, and while the magnitude of impact generally decreases as one moves away from a building or ground surface, the zone of influence can extend to hundreds of meters away from the building. To address and capitalize on these domain characteristics, this work integrates adaptations of existing object-centric neural implicit fields approaches. The scale and rotation are embedded as variables in the latent space through relating sampling bounds to a fixed bounding box across the entire building geometry's domain rather than individual elements. It additionally does not utilize absolute clamping of the prediction domain to learn the continuous field but instead adopt importance-based biasing terms to prioritize more relevant parts of the continuous domain. It explores ground-discretized latent embeddings as hybrid-neural representations to capitalize on the morphological characteristics of buildings and allow for the learning of the performance spatial hierarchies in the field around buildings. Lastly, it adopts shape-prior based learning approaches so that semantically-meaningful geometric parameters can be directly linked to performance outcomes, ensuring the model's explainability through direct parameter-to-performance associations, necessary for its usability in the design domain.

3.2. Geometry and performance learning framework

The proposed framework adopts a decoupled and sequential learning approach consisting of two modules that respectively encode the geometric representation and physical influence of buildings, as illustrated in Fig. 1. While the overall formulation is conceptually related to the globally modulated design introduced in the INFINITY framework [22], the approach presented generalizes beyond it in two fundamental ways: (1) replacing shift-based modulation with latent concatenation for conditioning for an architecture-agnostic conditioning mechanism, and (2) extending the framework to explicitly support spatially discretized latent representations in addition to global embeddings, thereby opening the design space to architectures that leverage spatial locality and hierarchical structure.

The geometric representation module adopts prior-based learning through auto-decoding. Initially randomized latent codes z_i'' for individual shapes are optimized concurrently with the network weights during training. At inference time, newly initialized codes are fitted to unseen shapes. These latent embeddings can take the form of either non-discretized vectors $z_{ND} \in \mathbb{R}^{1 \times n}$ as has been adopted by [64] or spatially ground-discretized tensors $z_{GD} \in \mathbb{R}^{s \times s \times n}$. This approach builds on earlier work [64] by the authors in which INRs were introduced for building geometry representation and performance-informed design. This paper instead develops INR models for performance fields, establishing an INR-based surrogate modeling framework rather than just geometry-only encodings. This work explores both variants for geometry and performance embeddings to assess the effectiveness of downstream architectures in exploiting locality and spatial hierarchies, motivated by the work of [79,83]. In all variants, geometry embeddings are globally-conditioned to enable the learning of geometry shape priors across all pipelines. The frozen geometry latent $z_{gi} \in \{z_{gND}, z_{gGD}\}$ is passed to a performance encoder $f_{enc,\phi}$, producing a performance embedding z_{pi} , which may be either global ($z_{pi} \in \mathbb{R}^{1 \times d}$) or spatially discretized ($z_{pi} \in \mathbb{R}^{s \times s \times d}$). The field decoder $f_{field,\psi}(z_i, x)$ predicts per-point physical quantities at query coordinates $x \in \mathbb{R}^3$. In the globally conditioned setting, z_i is concatenated directly with x . In the locally conditioned case, query points are first projected onto the ground plane using $\Pi(x)$, bilinear features are sampled from z_i , and the sampled features are concatenated with the vertical coordinate x_z before being fed into the decoder. This operation allows each 3D query to retrieve horizontally localized features from the ground-plane latent grid, while concatenation with x_z restores vertical dependence. It provides a memory-efficient conditioning scheme that preserves spatial coherence without requiring full volumetric embeddings.

Geometry Encoding:

$$z_{gi} = \arg \min_{z_{gi}} \mathcal{L}_g \left(f_{z_{gi}, \theta}(z_{gi}, x_j), y_{ij} \right) \quad (1)$$

$$z_{gi} \in \begin{cases} \mathbb{R}^{1 \times n} & (\text{non-discretized, } z_{gND}) \\ \mathbb{R}^{s \times s \times n} & (\text{ground-discretized, } z_{gGD}) \end{cases} \quad (2)$$

Performance Encoding:

$$z_{pi} = f_{enc,\phi}(z_{gi}) \quad (3)$$

$$z_{pi} \in \begin{cases} \mathbb{R}^{1 \times d} & (\text{non-discretized, } z_{pND}) \\ \mathbb{R}^{s \times s \times d} & (\text{ground-discretized, } z_{pGD}) \end{cases} \quad (4)$$

Field Decoding:

$$\hat{y}_{ij} = \begin{cases} f_{field,\psi}(z_i, x_j) & (\text{global conditioning}) \\ f_{field,\psi}(\text{bilinear}(z_i, \Pi(x_j)) \oplus x_{zj}) & (\text{local conditioning}) \end{cases} \quad (5)$$

3.3. Geometric learning approaches

There are two different schemes: G1, which is based on the use of non-discretized (ND) geometry embeddings and G2, which is based on ground-discretized (GD) ones.

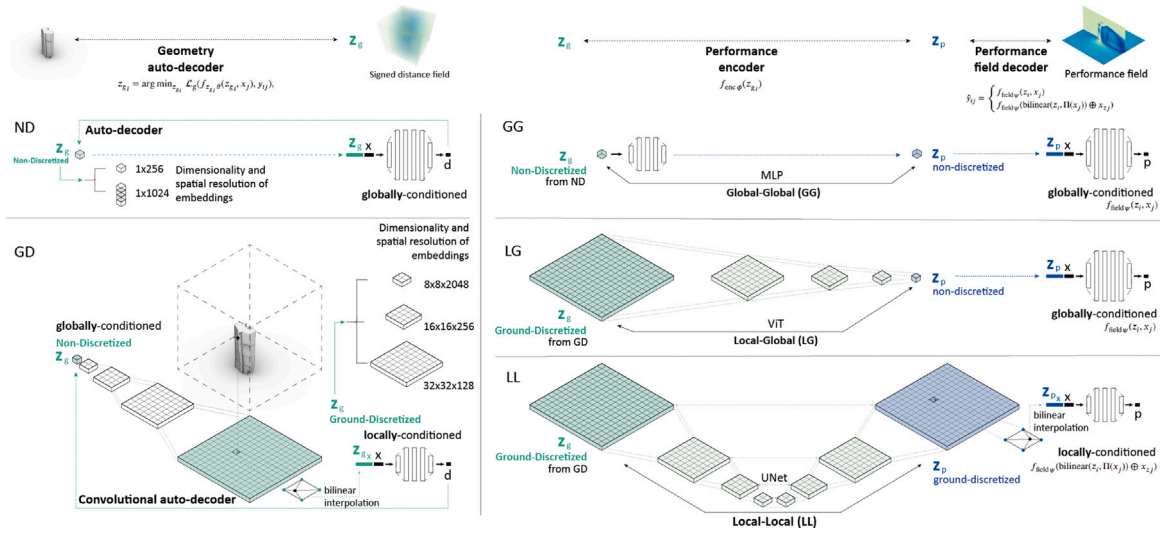


Fig. 1. Overview of the geometry and performance learning framework across proposed latent conditioning schemes.

3.3.1. ND geometry embedding

The non-discretized (ND) variant uses a Lipschitz-regularized DeepSDF network [2,96] with Gaussian Fourier feature coordinate mapping $\phi(x)$ [76] to learn an implicit signed distance function (SDF) mapping points $x \in \mathbb{R}^3$ to surface distances $d \in \mathbb{R}$. The Lipschitz regularization and Fourier feature mapping respectively contribute to the smoothness of the latent space and the ability of the model to capture high frequency details. The model is trained by minimizing a geometry loss L_g , combining a standard SDF loss, latent regularization [2], and Lipschitz regularization [96], where $\hat{f}_{ij} := f_\theta(z_i, \phi(x_j))$, and $\text{clp}(\cdot, \delta)$ denotes a clamped version of the distances:

$$L_g = \sum_{i=1}^N \sum_{j=1}^K \left[\alpha |\hat{f}_{ij} - d_j| + (1 - \alpha) |\text{clp}(\hat{f}_{ij}, \delta) - \text{clp}(d_j, \delta)| \right] + \gamma_z \sum_i \|z_i\|_2^2 + \gamma_l \prod_i \sigma^+(c_i) \quad (6)$$

The term α balances clamped and unclamped losses; δ defines the clamping threshold; γ_z and γ_l control the strength of latent and Lipschitz regularization, respectively; and c denotes per-layer Lipschitz coefficients. At test time, network weights are fixed and only the latent code z is optimized (refer to Appendix B for additional details).

3.3.2. GD geometry embedding

The globally-conditioned ground-discretized (GD) embedding is generated by upsampling a non-discretized (ND) latent into a 2D feature grid $s \times s \times n$ via a convolutional decoder, followed by a locally-conditioned MLP SDF decoder. This preserves shape prior learning while enabling spatial discretization [48,97]. The model is trained using a loss L_{gH} similar to L_g , replacing Lipschitz regularization with L2 weight decay $\gamma_w \sum_{i=1}^l \|w_i\|_2^2$, and substituting Lipschitz normalization with standard weight normalization (refer to Appendix B for additional details).

3.4. Performance learning approaches

Three performance learning schemes were explored: GG (global-to-global), LG (local-to-global), and LL (local-to-local). The three employ distinct architectures for the performance latent representation, followed by either a globally- or locally-conditioned field decoder. In GG, performance features are extracted via an MLP auto-encoder from frozen geometry latents; in LG and LL, they are obtained using vision architectures best suited for spatially discretized inputs like the GD latent grids: Vision Transformers (ViT) [98] and UNet [99], respectively

(refer to Appendix B.2 for additional details). All variants are trained to minimize a performance loss L_p comprising a reconstruction term and a weight regularization term:

$$L_p = \sum_{i=1}^N \sum_{j=1}^K \beta^{(d_j)} \omega^{(x_{z_j})} (f_\theta(z_i, \phi(x_j)) - p_j)^2 + \gamma_w \sum_{i=1}^l \|w_i\|_2^2 \quad (7)$$

This formulation introduces the exponential weighting terms β and ω , which prioritize learning near surfaces and near the ground, based on signed distances d and vertical coordinates x_z , respectively.

3.5. Evaluation framework

The evaluation framework establishes the experimental basis for assessing the proposed surrogate modeling pipelines, detailing the dataset, experimental setup, baselines, and performance metrics used for quantitative comparison.

3.5.1. Dataset

This work leverages the PRISM dataset [100], a large-scale synthetic geometry-to-performance dataset developed for the built environment. It includes 4759 procedurally generated building geometries, of which 831 are paired with high-resolution steady-state CFD simulations across 1–8 wind directions per model (average 7.6), totaling 4769 simulation instances. Mesh sizes for the full domains range from 16 to 38 million cells per building, with an average of 27 million. Sky view factor (SVF) calculations are provided for all 4759 geometries. Geometry samples are augmented through rotational transformations to support orientation-sensitive training. Train/validation/test splits are fixed at a ratio of about 70/10/20. Additional dataset details, characterizations, and a link to the full dataset are included in Appendix A of the supplementary material.

3.5.2. Experimental setup

A series of experiments were designed and conducted to evaluate the three instantiations (GG, LG and LL) of the proposed framework across multiple spatial discretization resolutions and latent dimensions, as detailed in Table 2. Latent resolutions were selected to maintain comparable parameter counts across models for a fair assessment of representational efficiency. Baseline encodings ($256 \times 256 \times 1$ and $256 \times 256 \times 2$) were matched to implicit configurations ($8 \times 8 \times 2048$, $16 \times 16 \times 256$, $32 \times 32 \times 128$), each within a similar range of 10^5 – 1.3×10^5 parameters. This setup allows evaluation of whether increased spatial discretization or higher latent dimensionality yields

Table 2Architectural configurations for GG, LG, and LL pipelines. N_θ values are in millions of parameters.

Model	Geo. Enc.	N_θ^g	Z_g	Perf. Enc.	N_θ^e	Perf. Enc. Arch.	Z_p	Perf. Dec.	N_θ^d	Perf. Dec. Arch.
GG ₁ × 256	MLP	11	1×256	MLP	0	$h_{\text{dim}} = [128, 64, 128]$	1×256	MLP	5	$h_l = 20, h_n = 512$
GG ₁ × 1024	MLP	46	1×1024	MLP	1	$h_{\text{dim}} = [512, 256, 512]$	1×1024	MLP	44	$h_l = 20, h_n = 1536$
LG ₈ × 8×2048	CNN+MLP	271	8×8×2048	ViT	63	$p_i = 2, n_i = 12$	1×768	MLP	63	$h_{\text{dim}} = [64, 64]$
LG ₁₆ × 16×256	CNN+MLP	12	16×16×256	ViT	58	$n_{hd} = 12, h_d = 768$	1×768	MLP	58	$h_{\text{dim}} = [256, 256]$
LG ₃₂ × 32×128	CNN+MLP	5	32×32×128	ViT	57	$mlp_d = 1536$	1×768	MLP	59	$h_{\text{dim}} = [1024, 1024]$
LL ₈ × 8×2048	CNN+MLP	271	8×8×2048	UNet	470	$n_d = 3, ngf = 1024$	8×8×2048	MLP	470	$h_{\text{dim}} = [64, 64]$
LL ₁₆ × 16×256	CNN+MLP	12	16×16×256	UNet	102	$n_d = 4, ngf = 256$	16×16×256	MLP	102	$h_{\text{dim}} = [256, 256]$
LL ₃₂ × 32×128	CNN+MLP	5	32×32×128	UNet	67	$n_d = 5, ngf = 128$	32×32×128	MLP	69	$h_{\text{dim}} = [1024, 1024]$

better performance within equivalent capacity. In global-conditioned FCN models, finer discretizations become computationally infeasible and less meaningful, motivating this balanced design. Unless otherwise specified, all models were trained in a supervised manner to predict per-point wind speed magnitudes using Adam optimizer with initial learning rates $\eta \in [10^{-3}, 10^{-4}]$, using a stepped decay schedule. Training continued until validation loss saturation, with early stopping after 10 epochs without improvement, typically 100–1000 epochs depending on architecture. In all experiments, geometry and performance models were trained separately end-to-end using mini-batches of sampled 3D query points, using fixed proportions of near-surface, uniform, and rejection samples to control the balance of interior, exterior, and boundary points. Performance quantities were normalized for numerical stability (wind speeds divided by 20, pressure coefficients scaled by 1/3), and all models used the same normalization and denormalization pipeline at training and evaluation. Geometry latents were initialized with Gaussian noise ($\sigma = 10^{-3}$ for ND, 10^{-4} for GD) and jointly optimized with model weights. Batch sizes ranged from 32 to 64 with 10^4 – 2×10^5 points per batch, depending on architecture capacity and memory footprint. MIT Supercloud [101] was used as the main computing resource for training and post processing. All the training experiments were conducted on Nvidia Volta V100 GPUs with RAM 32 GB running with an Intel Xeon Gold 6248 with 40 cores and 9 GB RAM per core, and all post-processing tasks including reconstruction, metrics, and visualizations were conducted on Intel Xeon Platinum 8260 processor with 48 CPUS and 192 GB total RAM (refer to Appendix B for additional experimental details).

3.5.3. Baselines

Two grid-based baselines are used for benchmarking at 256×256 input resolution: BL_{256×256×2}, which encodes global and local height maps as two channels (adapted from [39]), and BL_{256×256×1}, the widely adopted single-channel height-map surrogate [27–35]. Three-dimensional surrogate architectures such as CNN-, point-, mesh-, or transformer-based networks can, in principle, be applied to similar prediction tasks; however, their deployment becomes computationally intractable at the mesh and domain resolutions considered in this study (16–38M cells per building). These architectures operate on discretized volumetric or graph representations that require substantial downsampling to fit within GPU memory limits, leading to a loss of fine-scale geometric fidelity and degraded surface-level accuracy. Consequently, 2.5D grid-based frameworks remain the most computationally tractable and widely adopted alternatives for high-resolution surrogate modeling, offering a practical compromise between representational capacity and scalability despite their limited 3D awareness. The two defined grid-based baselines are trained using the generator UNet architecture of the pix2pix model [102], which remains the most representative configuration in this line of work. While variants such as adversarially trained UNets have been explored in prior work, the generator-only formulation is adopted here to provide a stable baseline without introducing GAN-related training instabilities. The same performance loss function used for the INR framework is employed for both baselines, excluding the biasing terms that are not directly compatible with slice-based prediction. These models are trained to predict physics quantities

on single horizontal slices of the flow domain. While this provides only partial field coverage, values at multiple vertical levels can be obtained by repeated queries at different slice heights.

3.5.4. Evaluation metrics

Geometry models are evaluated on unseen shapes using L1 distance and symmetric Chamfer distance normalized by bounding box diagonal. For performance models, 50th percentile absolute percentage error (p50-APE) and mean absolute error (MAE) are reported on two test splits: t_g , comprising geometries seen by the geometry module but unseen by the performance module; and t_c , consisting of geometries unseen by both. For each geometry, metrics are computed across three spatial categories: pedestrian-level points (3×3 m sampling), near-surface points (2×2 m sampling), and a random subset of 30,000 field points. Results are reported for points with positive SDF values (outside the geometry) within a 250-meter bounding box centered on each building. This filtering removes distortions from surface-transition artifacts and excludes trivial interior-zero regions. Metrics are aggregated per-geometry before averaging across the test set, and summarized with mean, standard deviation, and, where applicable, 50th and 90th percentiles. For geometric metrics (L1 and Chamfer), mean and standard deviation are computed after clipping at the 97th percentile to mitigate distortion from rare non-convergent cases, while percentiles are reported from the full distribution. No clipping was applied to performance metrics.

4. Results

The following results evaluate the proposed framework along four dimensions: geometry reconstruction quality, predictive accuracy across surface, ground, and field regions, comparison against grid-based baselines, and generalization to out-of-distribution (real-world) geometries.

4.1. Geometric reconstruction

Geometry reconstruction performance across ND and GD models is summarized in Table 3, which reports L1 and Chamfer distances on both t_g (a training subset, later used as a validation test set for the performance training) and t_c test sets. These results highlight the trade-offs between reconstruction fidelity, model complexity, and computational cost. The lowest reconstruction errors are achieved by the ND-based models, with lowest L1 of 0.7184 m for ND_{1×1024} and lowest Chamfer distance of 0.0076 m for ND_{1×256} on the held-out test set (t_c). While GD encodings achieve lower L1 losses, ND-based models consistently produce lower Chamfer distances, indicating superior global shape alignment. This improvement, however, incurs a substantial computational cost. For example, GD_{32×32×128} achieves L1 losses within 15% of ND_{1×256} while requiring only 50% of its parameters and achieving a 15-fold reduction in geometry training time. Increasing latent dimensionality by reducing spatial resolution, e.g., from GD_{32×32×128} to GD_{16×16×256}, results in higher reconstruction errors on the test set and longer training time. Further increasing the resolution to GD_{8×8×2048} yields only marginal performance gains,

Table 3

Geometry reconstruction performance on t_g and t_c sets for ND and GD models. Bold indicates best-in-class; underlining marks overall best.

Model	N_θ	$T_{g/e}$	Training data subset t_g [m]						Test set t_c [m]					
			μ_{L1}	σ_{L1}	$p_{90,L1}$	μ_{ch}	σ_{ch}	$p_{90,ch}$	μ_{L1}	σ_{L1}	$p_{90,L1}$	μ_{ch}	σ_{ch}	$p_{90,ch}$
ND ₁ × 256	11	16	0.6367	0.1167	0.8253	0.0047	0.0021	0.0076	0.7729	0.1835	1.033	0.0076	0.0057	0.0143
ND ₁ × 1024	46	35	0.5904	0.0971	0.7480	0.0062	0.0047	0.0123	0.7184	0.1382	0.9172	0.0112	0.0121	0.0234
GD ₈ × 8×2048	271	44	0.3205	0.1443	0.5357	0.0153	0.0045	0.0218	0.6957	0.3808	1.2544	0.0429	0.1066	0.0353
GD ₁₆ × 16×256	12	3	0.4133	0.2163	0.7511	0.0159	0.0046	0.0237	1.0432	0.6478	2.0612	0.4669	1.3514	1.2087
GD ₃₂ × 32×128	<u>5</u>	<u>1</u>	0.5124	0.2449	0.9023	0.0168	0.0046	0.0244	0.8291	0.5162	1.5568	0.0371	0.0770	0.0406

Table 4

Median absolute percentage error $p_{(50-APE)}$ across surface, ground, and field regions on t_g and t_c .

Model	N_θ	T_{tot}	Test set t_g $p_{(50-APE)}$ – unseen to perf. network						Test set t_c $p_{(50-APE)}$ – unseen to both networks					
			μ_{surf}	σ_{surf}	μ_{grd}	σ_{grd}	μ_{field}	σ_{field}	μ_{surf}	σ_{surf}	μ_{grd}	σ_{grd}	μ_{field}	σ_{field}
GG ₂₅₆	5	360	0.3693	0.1733	0.2628	0.0342	0.0583	0.0157	0.3785	0.1485	0.2929	0.0567	0.0848	0.0424
GG ₁₀₂₄	44	382	0.3035	0.0831	0.0381	0.0174	0.0236	0.0069	0.3593	0.1487	0.0688	0.0455	0.0453	0.0359
LG ₈ × 2048	63	47	0.3466	0.1033	0.1810	0.0485	0.0553	0.0099	0.4103	0.1325	0.2237	0.0936	0.0915	0.0467
LG ₁₆ × 256	58	43	0.3341	0.1188	0.1291	0.0298	0.0359	0.0100	0.3890	0.1344	0.1857	0.0703	0.0625	0.0418
LG ₃₂ × 128	59	115	0.3504	0.1782	0.0586	0.0128	0.0209	0.0072	0.3648	0.1428	0.0827	0.0570	0.0426	0.0439
LL ₈ × 2048	470	16	0.3434	0.0630	0.0900	0.0183	0.0334	0.0061	0.3997	0.0996	0.1096	0.0464	0.0556	0.0368
LL ₁₆ × 256	102	19	0.3690	0.0892	0.0829	0.0349	0.0244	0.0066	0.3945	0.1231	0.0909	0.0454	0.0485	0.0379
LL ₃₂ × 128	69	30	0.3582	0.1173	0.1158	0.0353	0.0210	0.0076	0.3705	0.1127	0.1480	0.0523	0.0454	0.0398

Table 5

Mean absolute error (MAE) in predicted values across surface, ground, and field regions on t_g and t_c .

Model	N_θ	T_{tot}	Test set t_g μ_{AE} [m/s] – unseen to perf. network						Test set t_c μ_{AE} [m/s] – unseen to both networks					
			μ_{surf}	σ_{surf}	μ_{grd}	σ_{grd}	μ_{field}	σ_{field}	μ_{surf}	σ_{surf}	μ_{grd}	σ_{grd}	μ_{field}	σ_{field}
GG ₂₅₆	5	360	1.4492	0.3320	0.6229	0.1210	0.5636	0.1550	1.6986	0.5467	0.7046	0.0965	0.7160	0.2695
GG ₁₀₂₄	44	382	1.2498	0.2925	0.1638	0.0429	0.3066	0.0798	1.7020	0.5050	0.2379	0.1132	0.4977	0.2571
LG ₈ × 2048	63	47	1.3907	0.4685	0.4752	0.1351	0.5149	0.1054	1.7362	0.5549	0.5114	0.1586	0.7079	0.2580
LG ₁₆ × 256	58	43	1.4019	0.5516	0.3924	0.0992	0.3898	0.1048	1.6785	0.6198	0.4716	0.1300	0.5676	0.2660
LG ₃₂ × 128	59	115	1.4443	0.7475	0.1958	0.0356	0.2981	0.0778	1.6138	0.6191	0.2573	0.1337	0.4707	0.3036
LL ₈ × 2048	470	16	1.3186	0.4103	0.2867	0.0644	0.3810	0.1082	1.7169	0.6499	0.3207	0.1040	0.5456	0.2499
LL ₁₆ × 256	102	19	1.4199	0.5178	0.2422	0.0516	0.3161	0.0929	1.6915	0.6485	0.2759	0.1043	0.4929	0.2477
LL ₃₂ × 128	69	30	1.4610	0.5806	0.2994	0.1167	0.3165	0.1334	1.6136	0.6549	0.3852	0.1270	0.4843	0.2972

with 0.6957 m L1 and 0.0429 m CD, despite a 50-fold increase in model parameters and a 44-fold increase in training time compared to GD_{32×32×128}, as shown in the N_θ and $T_{g/e}$ comparison in Table 3. Additionally, the GD_{32×32×128} model achieves the lowest error on the test set (t_c) despite showing the weakest training performance (t_g) among its counterparts, indicating improved generalization rather than reduced training loss.

4.2. Predictive abilities

Performance is evaluated against high-fidelity simulation outputs across multiple dimensions. This includes predictive accuracy across spatial regions, geometry–performance associations, the effects of importance-based biasing, and comparisons to grid-based baselines.

4.2.1. Performance predictions

Performance prediction results for ground and field points show strong accuracy, with average $p_{(50-APE)}$ values as low as 6% (corresponding to about 0.25 m/s and 0.5 m/s across ground and field) and under 12% (0.4 m/s and 0.7 m/s across ground and field) for the most challenging geometries. In contrast, near-surface predictions remain consistently challenging across models, with an average as low as 35% translating to deviations of about 1.6 m/s. This is partly due to the nature of APE, which disproportionately penalizes deviations when ground truth values are near zero, common in surface-adjacent regions where flow predictions can fluctuate rapidly between zero and nonzero, and due to the higher sensitivity of near-surface results to any deviations in geometric reconstruction. Tables 4 and 5 provide a comparative analysis of predictive accuracy across pipelines. While $p_{(50-APE)}$ captures

relative performance, MAE offers a more stable indication of absolute error, particularly near boundaries where target values approach zero. On the test set t_c , GD-based architectures (LG, LL) achieve comparable performance to ND-based GG models across surface and field points in both metrics, achieving similar/higher accuracy with significantly lower parameter counts and training time.

Increasing the latent dimension for ND encodings in the GG pipeline from 256 to 1024 improves accuracy, although at the cost of significantly longer training. Unlike the geometric reconstruction results, GD encodings offer substantial advantages over ND representations for field and surface predictions. They enable faster convergence, requiring only 1/30th the training time compared to the MLP-based GG models, and achieve comparable overall accuracy, as shown in Table 4. As shown in Fig. 2, increasing spatial discretization (and correspondingly reducing latent dimensionality) leads to significant improvements in predictive accuracy, a trend consistent across both LG_{VIT} and LL_{UNet} architectures. While LG and LL achieve similar quantitative results for surface and field results, LG_{VIT} shows better ground predictions than its LL_{UNet} counterparts. While this may favor the LG approach, the use of a spatially discretized embedding and locally-conditioned decoder in LL provides a structural advantage. This architectural choice may enable future extensions to multi-object settings such as urban blocks. Compositional modeling is not explored in this work, but the LL design is compatible with such scenarios. Fig. 3 visualizes prediction outputs and error maps for the best-performing model in each pipeline on the test sets t_g and t_c . These illustrate the models' ability to capture flow acceleration and deceleration zones around buildings of varying complexity. The largest errors are concentrated near surface boundaries and leeward regions.

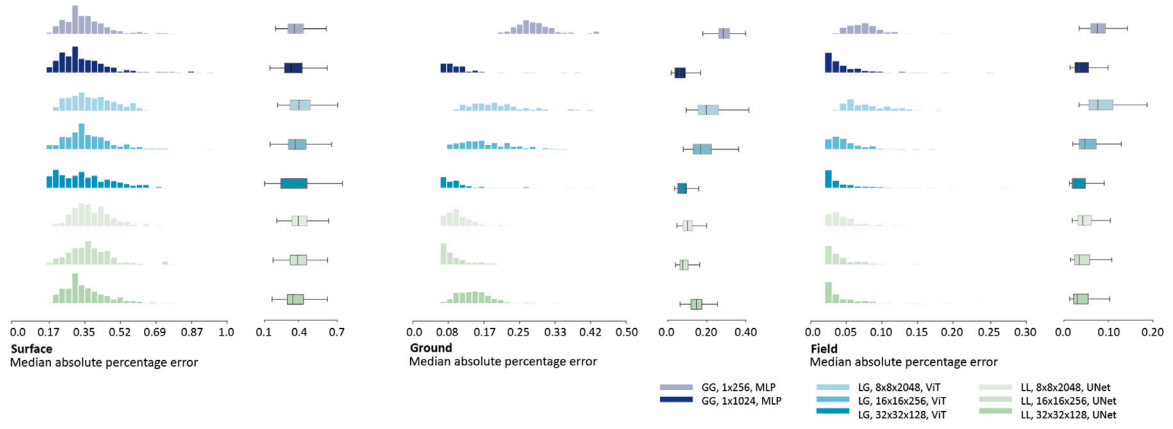


Fig. 2. APE distributions of performance loss across approaches and spatial discretization resolutions.

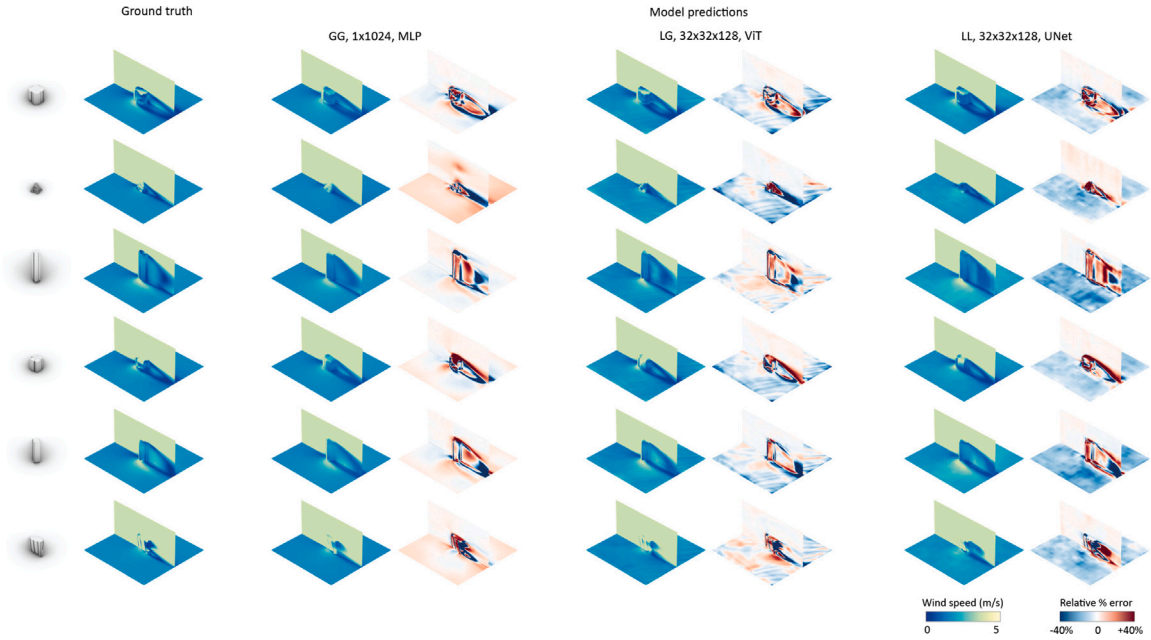


Fig. 3. Predicted wind flows and percentage error maps for geometries unseen to modules. Top three rows: t_g ; bottom three: t_c .

4.2.2. Geometry-to-performance associations

As the performance predictions are the product of the learned geometric reconstruction quality and the performance surrogate model capability, the relationship between both is investigated in Fig. 4. The average $p_{(50-APE)}$ for wind speed predictions at the ground level is plotted against the L1 loss and chamfer distance for all geometries in the test set t_c unseen by both networks for all experiments. The boundaries of the points across models are outlined to highlight the extents of their respective performance, the overlaps and distinctive edge conditions. The L1 loss and the chamfer distance plots provide a different perspective on the models' performance with respect to geometry. The $GG_{1 \times 1024}$ outline is characteristic of a much smaller plot footprint indicating a more compact distribution of errors in L1, but which are, for a large percentage of geometries, outperformed by the GL and LL approaches operating on GD geometry encodings. In terms of chamfer distance, GG approaches remain more powerful at globally representing the shapes. Across the three framework instantiations, the choice of spatial discretization and latent dimension have considerable implications on the potential lower and upper performance bounds as clear from the horizontal divisive boundary lines. The higher latent dimensions in $GG_{1 \times 1024}$ can at its lower point achieve $p_{(50-MAPE)}$ as low as 2.5% compared to 15% for $GG_{1 \times 256}$. Models $LL/GL_{32 \times 32 \times 128}$ achieve

better performance than the lower spatial discretization resolutions, with the larger impact observed in the case of the ViT in $GL_{32 \times 32 \times 128}$ where its best prediction is 3% $p_{(50-APE)}$ against 10% for $GL_{8 \times 8 \times 2048}$.

4.2.3. Importance-based biasing impacts

Varying levels of performance loss $p_{(50-MAPE)}$ were observed across domain regions: ground, surface, and field. To evaluate the effect of the surface-focused biasing term, a set of experiments varied the bias parameter β , which emphasizes learning near surfaces. As shown in Fig. 5, omitting the bias term significantly worsens near-surface predictions. Introducing a bias in the range $\beta \leq 10^{-4}$ reduces near-surface errors from 55% to 35% with minimal impact on ground and field accuracy. Beyond this range, near-surface gains saturate while ground and field losses increase.

Some performance applications are only (or mostly) concerned with surface-level predictions such as in the case of estimating the wind pressure coefficients for natural ventilation potential, and some metrics are uniform across most of the domain with variability only occurring near obstructions such as the sky view factor. To test the effectiveness of this approach, two $LL_{32 \times 32 \times 128}$ models were trained to predict pressure coefficients and sky view factor respectively using an extreme bias setting ($\beta \leq 10^{-32}$). Results in Fig. 6 show substantial surface-level

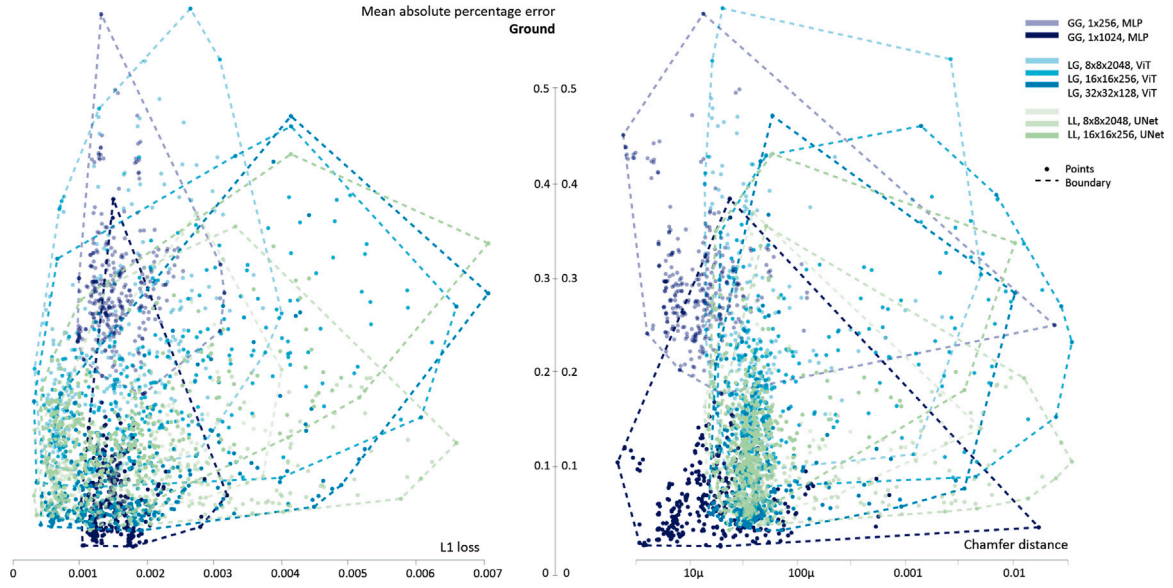


Fig. 4. Relationship between average wind speed $p_{(50-APE)}$ and geometric reconstruction losses (L1 loss and chamfer distance) on t_c .

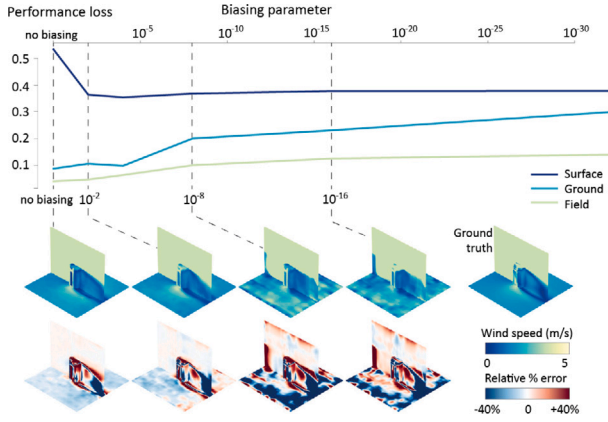


Fig. 5. Loss biasing term calibration impact on near-surface predictions as well as ground and field prediction trade-offs.

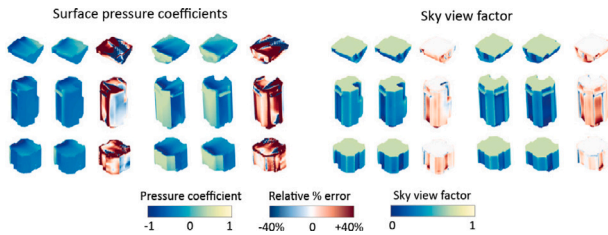


Fig. 6. Pressure coefficients and sky view factor predictions from $LL_{32 \times 32 \times 128}$ models trained using $\beta \leq 10^{-32}$.

improvements when field predictions are deprioritized, underscoring the importance of application-specific calibration.

4.2.4. Baseline comparisons

The proposed model is compared against two grid-based baselines (refer to Section 3.5.3). These baselines operate on horizontal slices of the domain and do not explicitly encode full 3D geometry, limiting predictions to the input resolution and preventing accurate surface or field-level outputs. Table 6 provides a cumulative quantitative evaluation of field performance across all test geometries, comparing

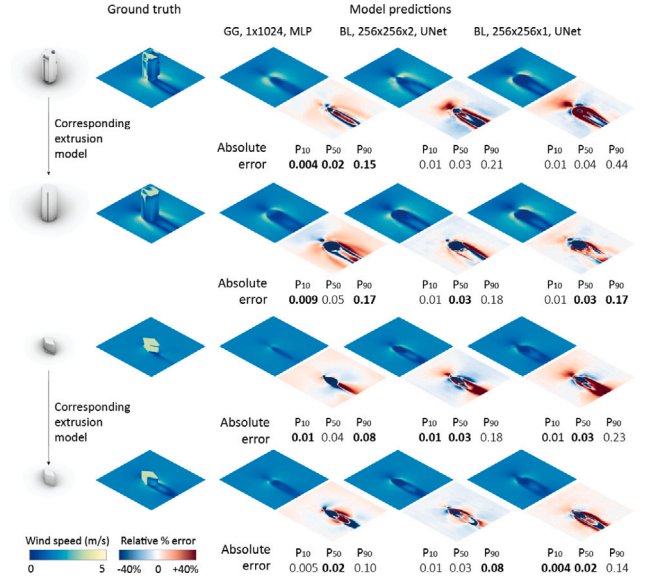


Fig. 7. Comparison of $GG_{1 \times 1024}$ predictions with baseline models for complex geometries and their corresponding extrusion models for two samples.

the proposed $LL_{32 \times 128}$ model against the two baselines ($BL_{256 \times 1}$ and $BL_{256 \times 2}$). Metrics are computed across horizontal slices at multiple heights and aggregated over all slices, as well as separated into 3D geometries and 2.5D extrusions. Across both test sets (t_g and t_c), the proposed model consistently outperforms baselines in terms of its distribution statistics.

While baseline performance is comparable on simplified extruded forms, accuracy degrades notably on more complex 3D geometries. $LL_{32 \times 128}$ demonstrates superior generalization in both settings, achieving lower errors and requiring significantly less training time. These results reinforce the need for models that leverage more expressive geometric representations, as well as architectures capable of capturing localized flow variations across diverse morphological conditions. Although some extruded test cases exhibit marginally lower baseline errors, these instances are not statistically representative of the overall trend observed across all test sets. To exemplify this observed pattern,

Table 6Field prediction accuracy of $LL_{32 \times 128}$ versus 2.5D baselines ($BL_{256 \times 1}$, $BL_{256 \times 2}$) on t_g and t_c .

Set	Model	N_θ	$T_{p_{tot}}$	All geometries				3D geometries				2.5D geometries			
				μ	σ	P_{50}	P_{90}	μ	σ	P_{50}	P_{90}	μ	σ	P_{50}	P_{90}
t_g	$LL_{32 \times 128}$	69	30	0.0267	0.0092	0.0246	0.0383	0.0284	0.0098	0.0263	0.0408	0.0245	0.0078	0.0231	0.0327
	$BL_{256 \times 2}$	54	258	0.0273	0.0152	0.0249	0.0481	0.0280	0.0161	0.0270	0.0496	0.0264	0.0137	0.0223	0.0462
	$BL_{256 \times 1}$	54	69	0.0371	0.0143	0.0350	0.0578	0.0380	0.0159	0.0374	0.0588	0.0358	0.0116	0.0338	0.0563
t_c	$LL_{32 \times 128}$	69	30	0.0544	0.0465	0.0374	0.1024	0.0565	0.0505	0.0380	0.1026	0.0516	0.0401	0.0367	0.1010
	$BL_{256 \times 2}$	54	258	0.0608	0.1022	0.0348	0.1116	0.0562	0.0727	0.0348	0.1085	0.0674	0.1311	0.0349	0.1139
	$BL_{256 \times 1}$	54	69	0.0637	0.0939	0.0436	0.1102	0.0625	0.0641	0.0446	0.1222	0.0655	0.1219	0.0416	0.1006

Table 7

Out-of-distribution test results using MAE and APE metrics across surface, ground, and field regions.

Model	$p_{(50-APE)}$									μ_{MAE} [m/s]								
	μ_{surf}	σ_{surf}	$\times_{t_c}$	μ_{grd}	σ_{grd}	$\times_{t_c}$	μ_{field}	σ_{field}	$\times_{t_c}$	μ_{surf}	σ_{surf}	$\times_{t_c}$	μ_{grd}	σ_{grd}	$\times_{t_c}$	μ_{field}	σ_{field}	$\times_{t_c}$
GG ₁₀₂₄	2.930	1.073	<u>x7.2</u>	0.044	0.020	x0.7	0.023	0.011	x0.5	2.414	0.666	<u>x1.4</u>	0.135	0.034	x0.6	0.318	0.109	x0.6
LG _{32 × 128}	2.822	1.444	<u>x7.1</u>	0.062	0.019	x0.7	0.016	0.006	x0.4	2.529	0.829	<u>x1.6</u>	0.170	0.046	x0.7	0.241	0.091	x0.5
LL _{32 × 128}	2.232	1.162	<u>x5.6</u>	0.090	0.030	x0.6	0.024	0.012	x0.5	2.094	0.765	<u>x1.3</u>	0.212	0.066	x0.6	0.317	0.145	x0.7

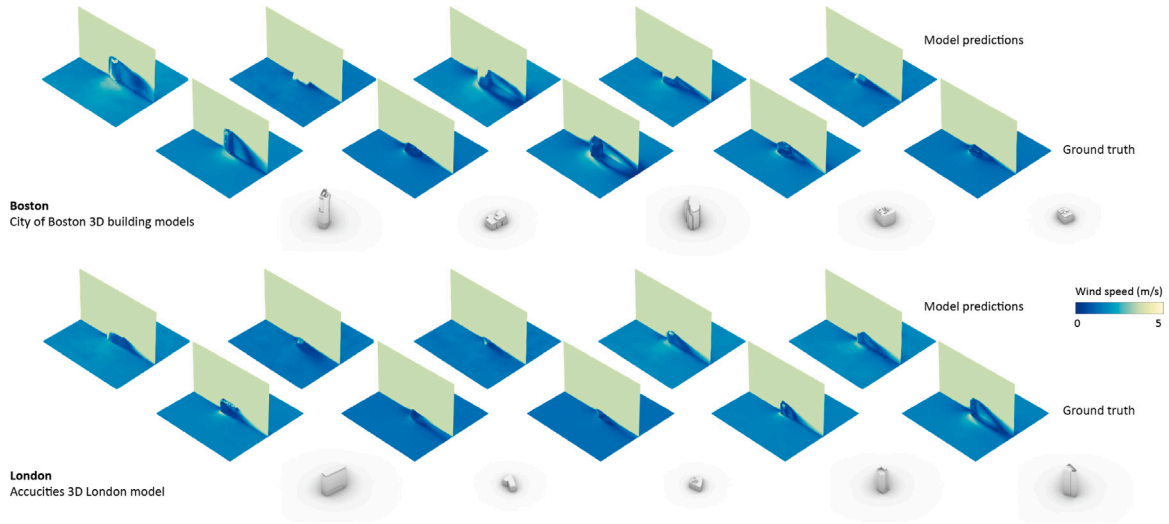
**Fig. 8.** Predicted wind speeds from GG_{1024} , $LG_{32 \times 32 \times 128}$, and $LL_{32 \times 32 \times 128}$, compared to ground truth CFD output on the out-of-distribution geometries.

Fig. 7 compares predictions from $GG_{1 \times 1024}$ and the baselines for two representative test cases. As shown in the flow predictions, while the baselines better capture flow patterns on simplified extrusions, their accuracy degrades on complex geometries, where the neural implicit model more effectively captures vertical and morphological variation.

4.3. Out-of-distribution generalization

To evaluate model generalization beyond the training distribution, performance was assessed on a real-world out-of-distribution (OOD) dataset comprising unseen building geometries and independently simulated CFD fields. The analysis focuses on predictive behavior across different spatial regions, surface, ground, and field, and quantifies deviations relative to in-distribution test performance.

4.3.1. Out-of-distribution evaluation dataset

The OOD dataset includes 10 buildings selected from two public sources: 6 from the Boston 3D Buildings dataset [103] and 4 from a 1 km² sample of London provided by AccuCities [104]. Candidate geometries were filtered for watertightness, LoD3+ complexity, and morphological diversity. High-resolution CFD simulations were performed for each geometry following the same preprocessing, meshing, and solver configuration used in the PRISM dataset. Simulations used

OpenFOAM [105] to solve steady-state RANS equations with a realizable $k-\epsilon$ model, in cylindrical domains scaled relative to building height and locally refined near surfaces. Runs used 32-core parallelization, with residual thresholds of 10^{-5} for velocity and 10^{-4} for pressure.

4.3.2. Generalization results

Model predictions from the GG, LG, and LL pipelines are compared against ground truth CFD simulation results for the OOD set, as shown in Fig. 8. Table 7 reports predictive accuracy using both the 50th percentile absolute percentage error $p_{(50-APE)}$ and the mean absolute error (MAE), alongside scaling factors quantifying the change in performance relative to the t_c test set. Results show strong generalization for ground and field predictions, with most models achieving lower or comparable errors relative to t_c (scaling factors $< \times 1.0$). These results imply that the training set may encompass a wider and more complex spectrum of geometries than those present in the OOD test set, which, while derived from real-world data, may exhibit less architectural variability than the synthetically constructed training distribution. In contrast, surface predictions degrade substantially across all models, with error scaling reaching $\times 5.6$ – $\times 7.2$. These elevated errors reflect persistent challenges in capturing near-wall behavior and interface discontinuities. This underscores the need for improved surface-aware strategies such as stronger near-boundary loss weighting, enhanced geometric encoding, or skewed sampling toward surface-adjacent regions. While

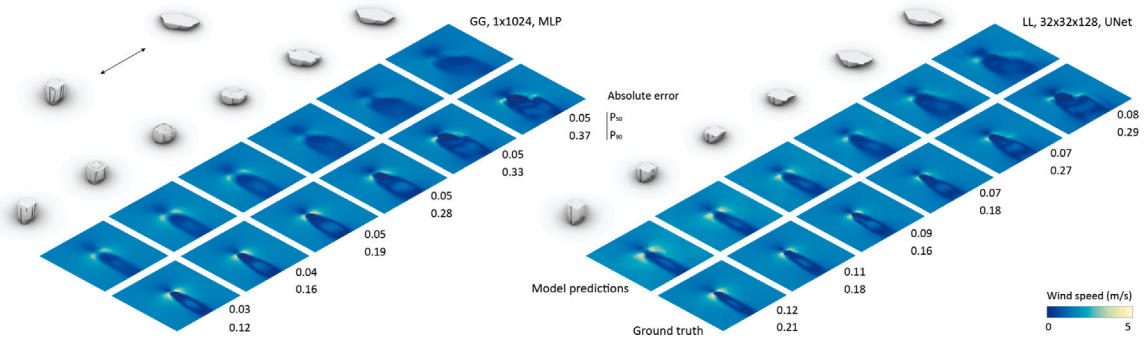


Fig. 9. Performance interpolations for a pair of trained geometries using the $GG_{1 \times 1024}$ and $LL_{32 \times 32 \times 128}$ models.

surface prediction remains an open challenge, the stability of ground and field performance highlights the potential of these models to extend reliably to real-world applications. Further evaluation on a broader range of real buildings would strengthen this finding, but such analysis remains limited by the scarcity of high-quality, simulation-ready datasets at scale.

4.4. Beyond surrogate modeling

The adoption of shape-conditioned geometry INR surrogate models extends the utility of the proposed framework beyond conventional surrogate modeling. By leveraging the learned continuous latent space of geometry, interpretable shape parameters can be directly linked to performance outcomes. This enables performance-informed design workflows, illustrated here through two applications: performance interpolation and latent space exploration. While performance-driven design space exploration has been an active area of research [64,106,107], the proposed framework provides a complementary pathway: it allows such exploration to be carried out on arbitrary geometries, and extends conditioning and evaluation beyond single aggregate attributes to spatially resolved full-field performance predictions.

4.4.1. Performance interpolation

Shape interpolations in the latent space between two or more elements, or more generally any latent arithmetic operations across shapes, provide the means to ideate and explore design possibilities. Directly connecting this ability to performance feedback can provide invaluable insights. Fig. 9 shows two examples of performance interpolations using the $GG_{1 \times 1024}$ and $LL_{32 \times 32 \times 128}$ models. Linear interpolation is performed on the geometry latent vectors of the selected building geometry pair and a forward pass through the geometry module enables their reconstruction. The generative ability enabled by the smoothness and continuity of the latent space is visible across both models, and the ability of the model to couple that with performance predictions is evident from the smoothness of the prediction variation as it moves closer from one end to the other. To validate the perceptual observations, CFD simulations were performed on the reconstructed interpolated meshes using Marching Cubes (MC) algorithm [108] at 128 resolution. $p_{(50-APE)}$ and $p_{(90-APE)}$ for each interpolation are recorded across both models. Results show that the performance interpolation enabled by the proposed framework are consistent to a large extent with the simulated values with a 3%–5% and 7%–12% $p_{(50-MAPE)}$ deviation respectively for each model.

4.4.2. Performance latent space

By compressing higher dimensional spaces into reduced dimensions, the continuous latent space enables capturing the underlying structure, key features and structural similarities characteristic of the design space. Clustering geometry latent dimensions facilitates the identification of shape groupings with similar morphologies. The introduction of

the performance encoder module in this surrogate modeling framework transforms geometry encodings into compact performance encodings. This learned transformation repositions shapes in the latent space from groupings of morphologies to groupings of performance impacts. Fig. 10 shows the results of k-means clustering (8 classes) applied to the trained geometry and performance encodings of the $GG_{1 \times 1024}$ model. The clusters are visualized through a t-SNE plot, a selection of buildings within (and close to the centroid of) each cluster and the cumulative probability distribution (CDF) of ground-level wind speeds for all geometries. On one hand, it can be observed that geometry embedding groupings are inherently groupings of shapes that have morphological similarities, e.g. shape, scale, dimension, orientation, etc. and while this may overlap with similar performances, it is not always the case. On the other hand, performance embedding clusters show substantially different morphologies that only have their wind impact flow in common. The same is evident in the CDF plots of performance embeddings, where buildings with similar performance functions correspond more closely to their respective clusters, in contrast with their geometry embedding counterparts. The ability to identify geometries within a wide spectrum of scale, morphology and complexity that hold similar performance impacts in a complex physical domain like wind flow exemplifies one of the key benefits of such the proposed learning framework.

5. Discussion

The results collectively illustrate how implicit neural representations can serve as compact, geometry-aware surrogates for building-scale performance modeling. Beyond their quantitative performance, these findings clarify where the approach succeeds, where it remains limited, and how it can extend toward more complex and performance-driven design contexts.

5.1. Insights and impact

The findings of this study demonstrate that INRs provide a viable and competitive approach for surrogate modeling in the built environment, supporting both field- and surface-level predictions while also enabling shape-prior-informed and performance-driven design exploration. Several actionable insights emerge. First, latent discretization choice matters: ND encodings offer the best global shape alignment (lowest Chamfer distances), while GD encodings achieve comparable L1 accuracy with far fewer parameters and significantly lower training time. Second, predictive accuracy can be improved by increasing spatial discretization and reducing latent dimensionality, with GD-based LG and LL pipelines matching GG baselines at substantially reduced computational cost. Third, near-surface predictions remain the most difficult, but importance-based biasing offers a practical means of reducing error without compromising ground and field accuracy, and extreme biasing is effective when surface-specific metrics (e.g. pressure

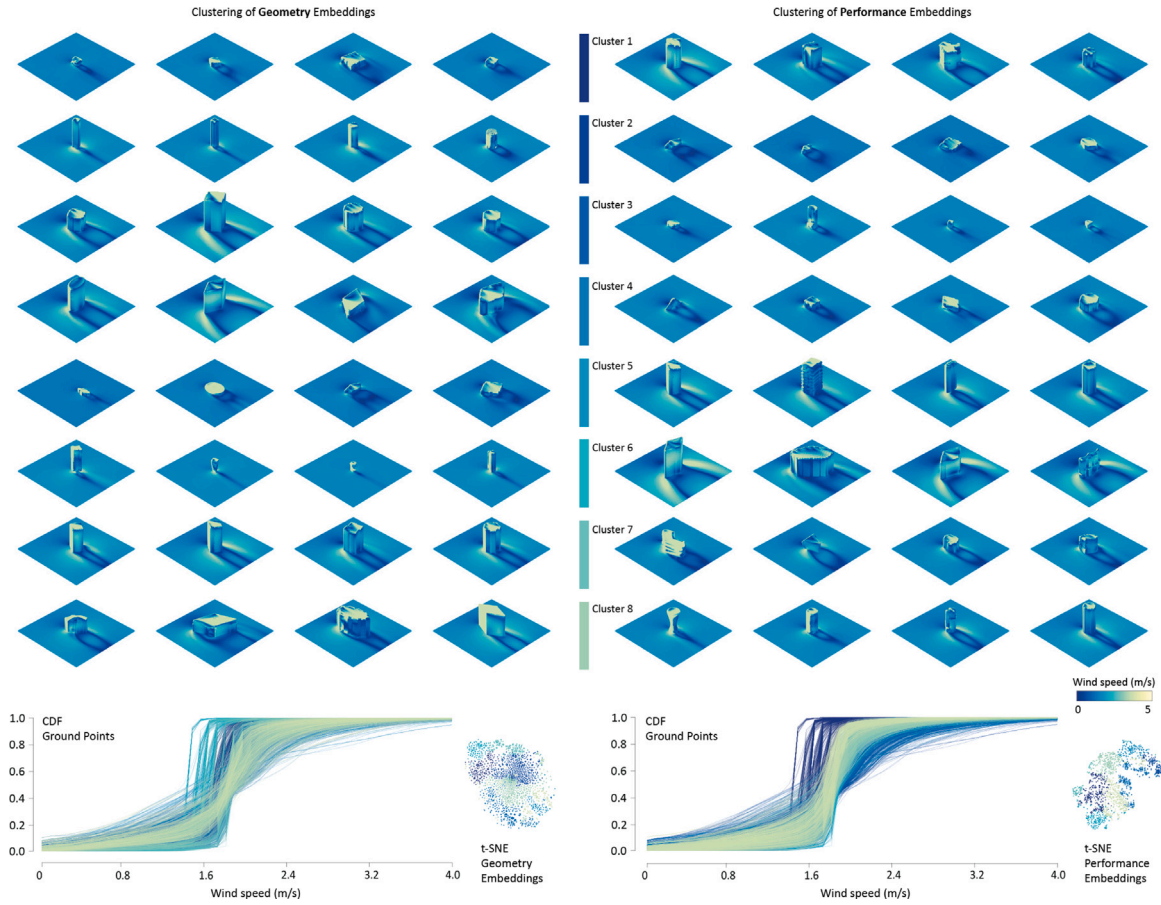


Fig. 10. K-means clustering on the trained geometry and performance encodings of the $GG_{1 \times 1024}$.

coefficients, sky view factor) are the primary target. This limitation is not intrinsic to the INR formulation itself but stems from the reduced sampling density and representational capacity at the solid-fluid interface, where gradients are steepest. Increasing near-surface sampling resolution and expanding geometry embedding capacity are expected to further improve performance in future extensions. Fourth, compared with conventional grid-based baselines, INR surrogates generalize more effectively to complex 3D morphologies, highlighting the value of expressive geometric encodings. Finally, out-of-distribution experiments confirm that ground and field predictions remain stable or improve relative to in-distribution, reinforcing the generalization potential and applicability of this framework for early-stage performance inference. Taken together, these findings address the three guiding questions introduced at the outset. The results show that implicit neural representations provide a tractable and scalable formulation for building-scale physics, that the learned latent structure captures coupled geometric and physical relationships for joint field and surface prediction, and that the resulting representation functions as a shape-informed prior that supports performance-guided exploration and generalization across diverse morphologies.

5.2. Scope and applicability

The presented surrogate modeling framework is formulated for isolated single-building domains, an abstraction that omits contextual geometry while enabling controlled identification of morphology-performance relationships. Within this scope, it is applicable to building-scale predictive tasks such as estimating surface pressure coefficients, surrounding wind flow, and solar access, particularly in early design scenarios where the surrounding context is fixed and only the target geometry varies. The latent embedding learned under this formulation

functions as a structured shape prior, supporting semantically consistent interpolation, clustering, and exploration of latent directions correlated with performance outputs. This property enables both optimization and inverse design based on interpretable geometry-performance trade-offs. Contextual effects can be approximated through boundary-aware constraints (e.g., occlusion masks or directional wind exposure conditions) without requiring explicit multi-building simulations. In many urban settings, however, inter-building interactions such as wake dynamics and mutual sheltering dominate performance and must be captured explicitly. To enable such extensions, the locally-conditioned LL pipeline was designed to be compositionally extensible, providing a foundation for block- and neighborhood-scale surrogates in future work. While the model was trained under fixed boundary conditions, specifically an inlet wind speed of 5 m/s at a 10 m reference height and a terrain roughness length of 1.0, variations in wind conditions can be approximated through logarithmic wind-profile scaling or directly incorporated as conditioning parameters within the surrogate framework. Material or surface heterogeneity, though not explicitly modeled in the current setup, could likewise be introduced by augmenting the geometric encoding with additional property channels, similar to how NeRF-based [109] methods jointly represent density and radiance.

5.3. Limitations and opportunities

The framework exhibits several limitations that highlight avenues for refinement and extension. Although it captures a wide range of morphological variation, fine-grained geometric details remain challenging under the globally-conditioned setting, highlighting the need for improved trade-offs between reconstruction fidelity, latent space smoothness, and computational cost. Generalization is also limited by dataset scale and diversity as producing large geometry-to-performance

datasets through high-fidelity simulation remains computationally expensive, and unlike other fields, open-source resources remain scarce. While importance-based biasing improves surface-level accuracy, this may be insufficient for applications where field-level fidelity cannot be compromised. Moreover, the INR formulation requires latent optimization at inference to estimate the geometry code for each new test instance, which makes it not fully end-to-end in contrast to purely feed-forward surrogates and can result in higher inference times (which could be partially mitigated through meta-learning initializations). The surrogate model's accuracy in this framework is as well jointly determined by both the geometry encoding and the performance field prediction, introducing a cumulative error structure that differs from explicit-encoding surrogates where geometric input is directly provided. Improving disentanglement between geometry and performance components thus remains an important avenue for future work. Additionally, applying models trained on synthetic data to real-world scenarios requires careful validation and adaptation to application-specific contexts. Robust and generalizable surrogate models will emerge from coupling large-scale and diverse datasets, with formulations that better resolve surface and field dynamics, capture contextual interactions, and incorporate application-driven validation.

6. Conclusion

Through the introduction of an INR surrogate modeling framework, the paper demonstrates the capacity of a shape-conditioned neural implicit field to act as an effective compact representation and functional input for downstream inference of physical fields in the built environment. The framework is evaluated on diverse geometric conditions to assess its predictive power and generalization capacity. The results show competitive predictive abilities to the predominant image-to-image translation operating on 2.5D geometric encoding for complex geometries and in par performance on simple geometries and extrusion models. In addition to expanding the repertoire of handled geometric complexities, the proposed approach expands the model's predictive field from single planes to field predictions. Importance-based biasing terms are proposed to prioritize the learning of more relevant parts of the domain, which are critical for applications where surface-level predictions are needed. The paper discusses three distinct geometry-to-performance pipelines distinguished by the types of latent conditioning used for the geometry and performance modules. It shows that operating on hybrid-ground geometry representations can be more effective at capturing spatial performance field hierarchies while reducing the computational overhead. It highlights the ability of locally conditioned performance field decoders to produce consistent predictions as a considerable advantage because it reflects an inherent ability to scale the performance surrogate models to urban blocks (multi-object settings) which are more representative of real-world environments. Performance-informed design applications extending beyond surrogate modeling and enabled by the proposed shape prior based learning approach are discussed and include performance-based categorization of building morphologies and performance-informed ideation. Future work can investigate how this framework can capitalize on its modality- and discretization-independent nature to handle multiple modalities (3D, 2D sketch, text) and multi-disciplinary performance measures (physical, geometric, implicit, aesthetic). Within the broader context of automation in construction, the framework underscores how physics surrogate models can enhance efficiency in design and engineering processes, accelerate performance-informed ideation and conceptualization of built environments, and ultimately contribute to more streamlined, data-driven, and sustainable design-to-construction workflows.

CRediT authorship contribution statement

Sarah Mokhtar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Caitlin Mueller:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Datasets

All experiments are conducted using the PRISM dataset [100], a large-scale, synthetic geometry-to-performance benchmark tailored to the built environment. PRISM comprises 4759 procedurally generated building geometries, including both unique geometries and their matched extrusions. A subset of 831 geometries is paired with high-resolution CFD simulations spanning 1 to 8 cardinal wind directions. Each geometry is provided in multiple modalities (e.g., watertight surface meshes, point clouds, structured descriptors) and includes aligned velocity, pressure, and sky view factor outputs.

Geometry Learning Dataset: Experiments on geometric representation learning use fixed training and test partitions containing 33936 and 4712 samples, respectively, augmented via axis-aligned rotations of point coordinates while preserving signed distance function (SDF) labels. This formulation enables rotation-variant representations, which are critical for tasks involving orientation-dependent phenomena (e.g., wind and solar interaction). Supervision is point-based, using SDF regression on sampled point–SDF pairs. Points are drawn from PRISM's predefined rejection, near-surface, uniform, and surface samplings at a 1:1:0.1:1 ratio. During training, the aggregated set is sub-sampled to maintain a 0.4:0.4:0.2 ratio across positive, negative, and zero-SDF categories.

Performance Learning Dataset: Unless otherwise noted, models are trained to predict wind speed magnitudes. Performance learning adopts fixed partitions of 2720 training, 72 validation, and 330 test instances. The validation set (t_g) contains geometries not seen during performance model training, while the test set (t_c) comprises geometries excluded from both geometry and performance training. Each training sample consists of a 3D point coordinate, its associated SDF value (used exclusively for the surface importance-weighted loss term), and a scalar performance label (e.g., wind speed), extracted from OpenFOAM simulation outputs via the SAMPLE utility using the cell–Point interpolation method. Performance quantities are normalized to the [0, 1] range for numerical stability, with wind speed values divided by 20 and pressure coefficients scaled by a factor of 1/3.

Performance training points are drawn from the rejection, near-surface, uniform, and domain-level horizontal slice sampling sets, and are sub-sampled to enforce a 0.9:0.1 ratio between positive and negative SDF values. In addition to the point sets provided in PRISM, three supplementary sampling strategies are introduced:

- *Domain horizontal slices:* Points sampled on horizontal planes at 512×512 resolution with 3 m spacing, at the following heights: 1.5, 3, 4.5, 6, 7.5, 9, 10.5, 12, 13.5, 15, 30, 60, 120, 240, and 480 m.
- *Domain vertical slices:* Points sampled on vertical planes aligned with the N–S, E–W, NE–SW, and NW–SE directions, each at 512×512 resolution with 3 m spacing.
- *Performance surface sampling:* Building surface points sampled on a 2×2 m grid with a 0.5 m offset from the façade.

Of these, the domain vertical slices and performance surface sampling sets are used exclusively for evaluation, testing, and visualization purposes. Only the domain horizontal slices are incorporated into the training set.

Table 8
ND model hyperparameters (parameters that differ across experiments).

ID	z	σ_{z_0}	h_l	h_n	h_s	lr_{m_0}	lr_{z_0}	γ_z	γ_l	b/n_s
ND1	256	0.001	8	512	4	0.001	0.0032	10^{-5}	10^{-14}	64/20480
ND2	1024	0.001	12	512–2048	4	0.001	0.0032	10^{-3}	10^{-20}	32/16380

Table 9
GD model hyperparameters (parameters that vary across experiments).

ID	z	z_{GD}	σ_{z_0}	$\sigma_{z_{GD_0}}$	c_l	h_n	lr_{m_0}	lr_{z_0}	lr_{sd}	lr_{si}	γ_z	γ_l	b/n_s
GD1	2048	$8 \times 8 \times 2048$	0.1	10^{-4}	3	2048	0.0001	0.00064	0.5	250	10^{-5}	10^{-1}	32/24576
GD2	256	$16 \times 16 \times 256$	0.1	10^{-4}	4	256	0.001	0.0016	0.8	50	10^{-3}	10^{-4}	64/100000
GD3	128	$32 \times 32 \times 128$	0.1	10^{-4}	5	128	0.001	0.0016	0.5	250	10^1	10^{-4}	64/200000
GD4	8	$128 \times 128 \times 8$	0.1	10^{-4}	7	8	0.0001	0.00064	0.5	250	10^{-3}	10^{-5}	64/240000

Appendix B. Implementation details

B.1. Geometric representation learning

Two types of globally conditioned neural implicit SDF models were explored: a non-discretized (ND) latent vector and a ground-discretized (GD) latent grid. All models were trained for 1000 epochs, except GD4, which was stopped at 600 epochs due to computational constraints.

Glossary of architectural parameters used in Tables 8 and 9: z : latent dimension, $\sigma_{z_0} / \sigma_{z_{GD_0}}$: latent initialization std. dev., z_{GD} : spatial latent grid shape, h_l : number of decoder layers, h_n : number of hidden units, h_s : skip connection interval, c_l : number of transpose convolutions, lr_{m_0} / lr_{z_0} : initial learning rates for model/latent, lr_{sd} / lr_{si} : learning rate decay factor/interval, γ_z / γ_l : latent/Lipschitz regularization weights, d : dropout rate, b : batch size, n_s : number of point samples per batch, ϕ : Fourier feature mapping size and scale, nl / nl_o : nonlinearity (hidden/output), w_n : weight normalization type, α : clamped/unclamped loss weighting, δ : surface clamp distance.

Non-discretized (ND) Geometry Embedding: ND models use a global latent vector $z \in \mathbb{R}^d$ and a Lipschitz-regularized DeepSDF decoder [2,96] with Fourier feature coordinate encoding [76]. The decoder is a multi-layer perceptron (MLP) with skip connections every h_s layers, hidden layer width h_n , and h_l total layers. Fourier encoding uses $\phi = (64, 1)$, corresponding to 64 features with Gaussian scale 1. All ND models use a linear activation in hidden layers and ReLU at the output, with Lipschitz normalization and dropout $d = 0.2$.

The loss combines clamped and unclamped SDF losses, weighted by $\alpha = 0.1$ with a clamp distance $\delta = 0.001$. Learning rate decays by a factor lr_{sd} every lr_{si} epochs. The optimizer regularizes both latent codes (γ_z) and Lipschitz constants (γ_l). Parameters shared across ND experiments: $\phi = (64, 1)$, nl = Linear, nl_o = ReLU, w_n = Lipschitz, $d = 0.2$, $\alpha = 0.1$, $\delta = 0.001$, $lr_{sd} = 0.5$, $lr_{si} = 200$.

Ground-discretized (GD) Geometry Embedding: GD models use a spatial latent grid $z_{GD} \in \mathbb{R}^{s \times s \times c}$ decoded through c_l transpose convolutions (kernel size $k = 3$, stride $st = 2$, padding $p = 1$). The representation network uses LeakyReLU activations with linear output; the field decoder uses ReLU with a linear output. Dropout is 0 for all GD variants. Coordinate Fourier features are not used. The SDF loss, learning rate decay schedule, and regularization terms mirror the ND configuration. GD4 did not converge across all training geometries, likely due to excessive spatial discretization with under-parameterized channel dimension.

B.2. Performance learning

Three performance learning schemes were explored: GG (global-to-global), LG (local-to-global), and LL (local-to-local). Each variant uses a performance encoder to produce a latent representation Z_p , conditioned either globally or locally, which is then decoded by an MLP to reconstruct continuous spatial fields. The architectural configurations

for all experiments are detailed in Table 2. All models are trained to minimize the same performance loss L_p introduced in the main paper, which includes a reconstruction term weighted by surface and ground proximity terms $\beta^{(d)}$ and $\omega^{(x_z)}$, and an ℓ_2 regularization term on decoder weights.

- **GG (global-global):** In GG models, geometry latents $Z_g \in \mathbb{R}^{1 \times d}$ are first frozen. A shallow MLP auto-encoder is used to encode performance information into $Z_p \in \mathbb{R}^{1 \times d}$. The encoder uses LeakyReLU (0.2) activations, dropout (0.2) for all layers except the last, weight normalization, and skip connections every 2 layers. The MLP field decoder uses ReLU activations, dropout (0.5), and skip connections every 4 layers.
- **LG (local-global):** In the LG models, performance encoding is learned directly from the spatial latent grid using a Vision Transformer (ViT) [98], adapted from torchvision's standard implementation [110]. For different resolutions, ViT configurations vary in patch size (p_s), number of transformer layers (n_l), attention heads (n_{hd}), hidden size (h_d), and MLP dimensions (mlp_d). The decoder remains an MLP that uses FFT-based coordinate encoding ($\phi = (64, 1)$), ReLU nonlinearity, dropout (0.2), and weight normalization.
- **LL (local-local):** In LL models, both the geometry and performance representations are spatially discretized. Performance encoding is performed using a UNet backbone adapted from the Pix2Pix generator [102] to accommodate lower spatial resolutions. Key parameters include the number of downsampling blocks (n_d) and number of filters in the last convolutional layer (ngf). The decoder follows the same architecture as in the LG scheme.

Glossary of architectural parameters used in Table 2: h_{dim} : hidden layer dimensions of MLP decoder; h_l : number of hidden layers; h_n : hidden layer width; p_s : ViT patch size; n_l : number of ViT transformer layers; n_{hd} : number of attention heads; h_d : ViT hidden dimension; mlp_d : MLP dimension inside transformer blocks; n_d : number of downsampling stages in UNet; ngf : number of filters in last UNet conv layer; Z_g , Z_p : geometry and performance latent dimensions.

Data availability

All data, code, and instructions necessary to reproduce the experiments and results reported in this paper are available at <https://github.com/samokhtar/PerFORM>.

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